

CO2xCOMM: Porewater Methane

July 2026 Samples

2026-08-28

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##Run Information

```
cat("Run Information: Brenden D. Blakley ") #lets you know what section you're in
```

Run Information: Brenden D. Blakley

```
library(here)
```

here() starts at C:/Users/blakleybd/OneDrive - Smithsonian Institution/Documents/GitHub/GCREW

```
#set the run date & user name
```

```
sample_year <- "2026"
```

```
sample_month <- "July"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month
```

```
GHG_files <- c("Raw Data/20260821_GCREW_2M_1M_formatted.csv")
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
log_path <- here("Porewater QAQC Logs", "GCREW_CH4_QAQC_Log.csv")
```

```
# Define final path for data be sure to change year and month and run number
```

```
final_path <- "Processed Data/GCREW_CO2xCOMM_Porewater_CH4_202607.csv"
```

```
#Gas in Water Calculations
```

```
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
```

```
TempC = 25 #Equilibration Temperature (Celsius)
```

```
volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
```

```
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g)
```

```
#record any notes about the run or anything other info here:
```

```
run_notes <- ""
```

```
#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"
```

```
cat(run_notes)
```

##Set Up

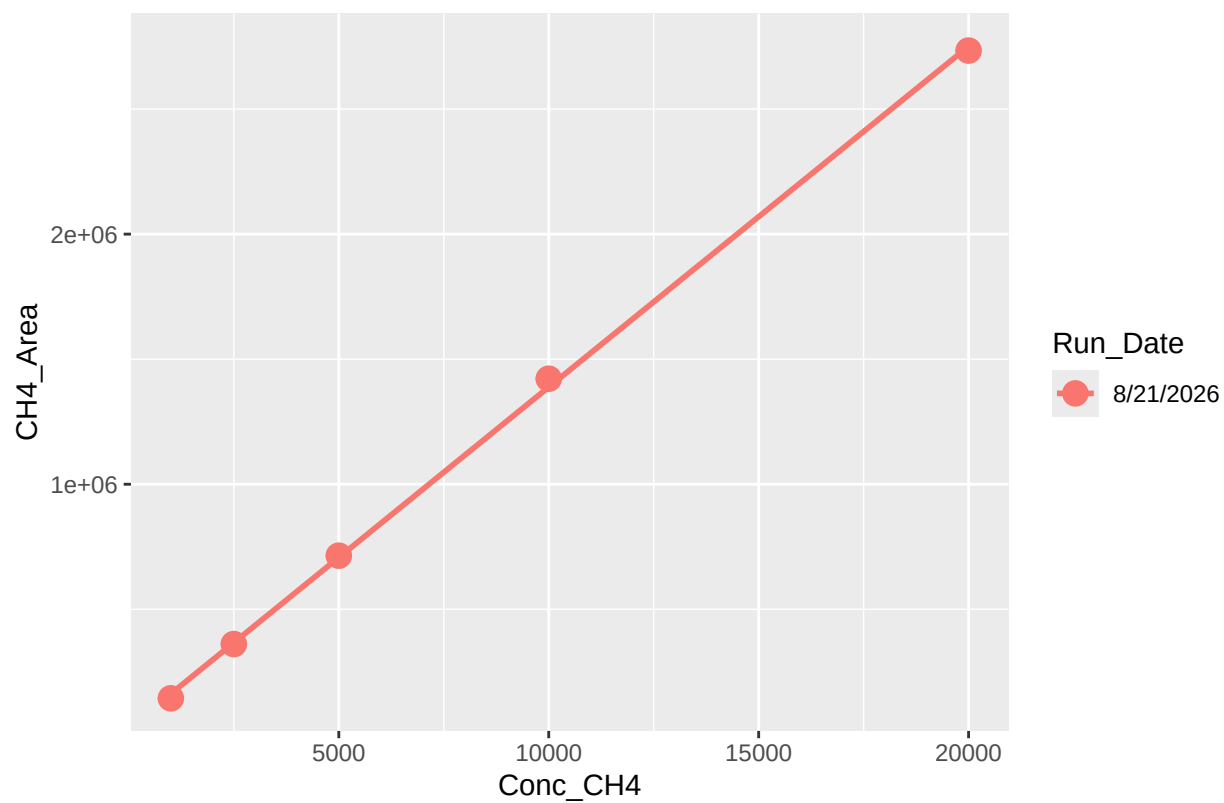
##Read in metadata and create similar sample IDs for matching to samples

1 Read in data files

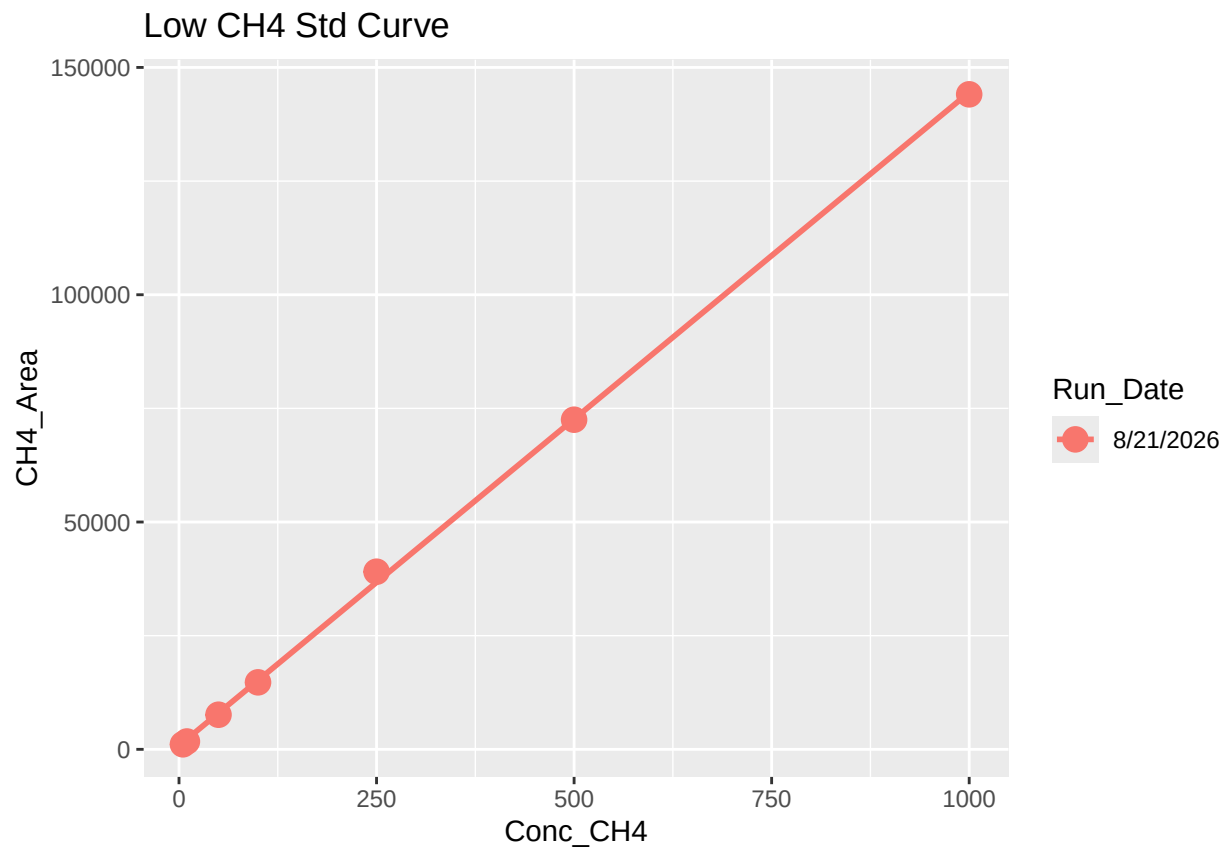
2 Assess standard curves

'geom_smooth()' using formula = 'y ~ x'

High CH4 Std Curve

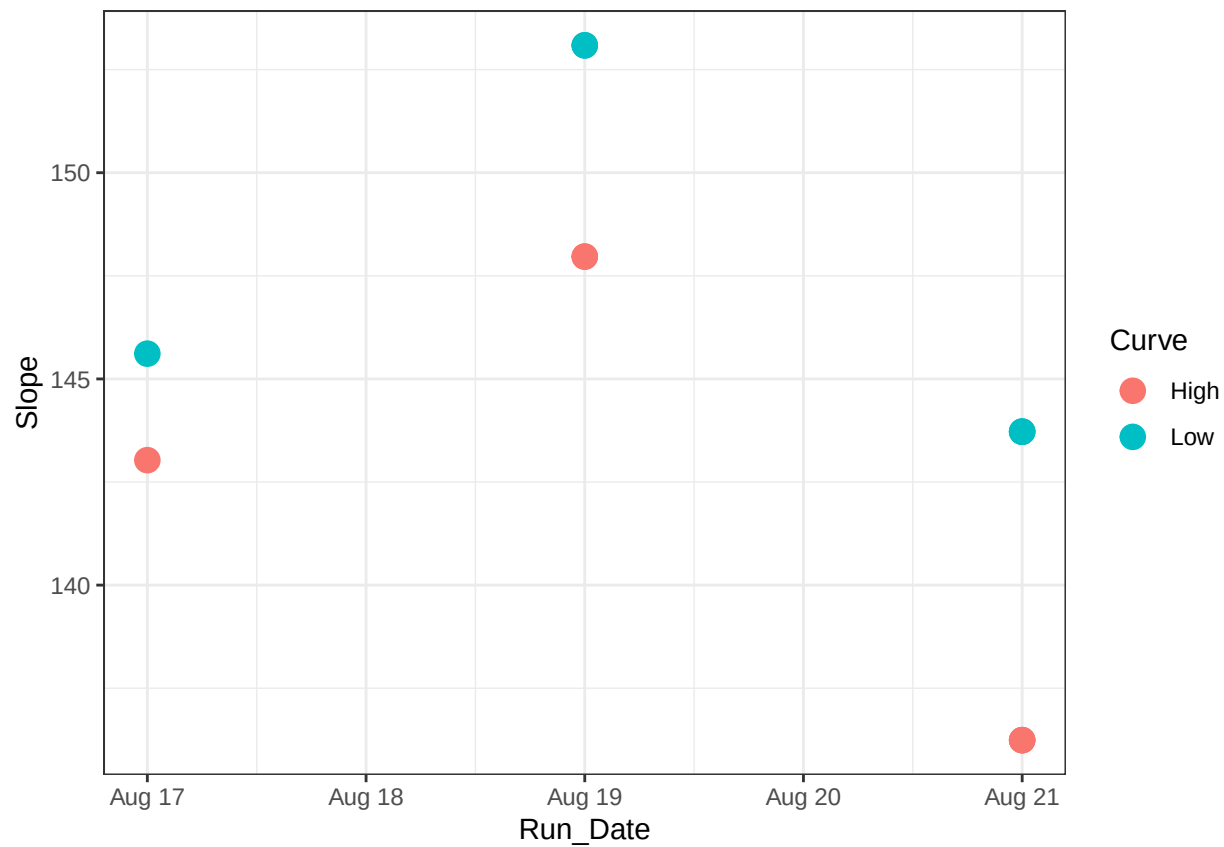


```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## Joining with 'by = join_by(Run_Date)'
```

##	Curve	R2	Slope	Intercept	Run_Date
## 1	High	0.9994458	136.2406	25980.5484	8/21/2026
## 2	Low	0.9995311	143.7214	801.9223	8/21/2026
## 3	High	0.9999168	147.9666	17873.3851	8/19/2026
## 4	Low	0.9997966	153.0888	407.2899	8/19/2026
## 5	High	0.9999646	143.0303	7587.2742	8/17/2026
## 6	Low	0.9996977	145.6125	745.1636	8/17/2026



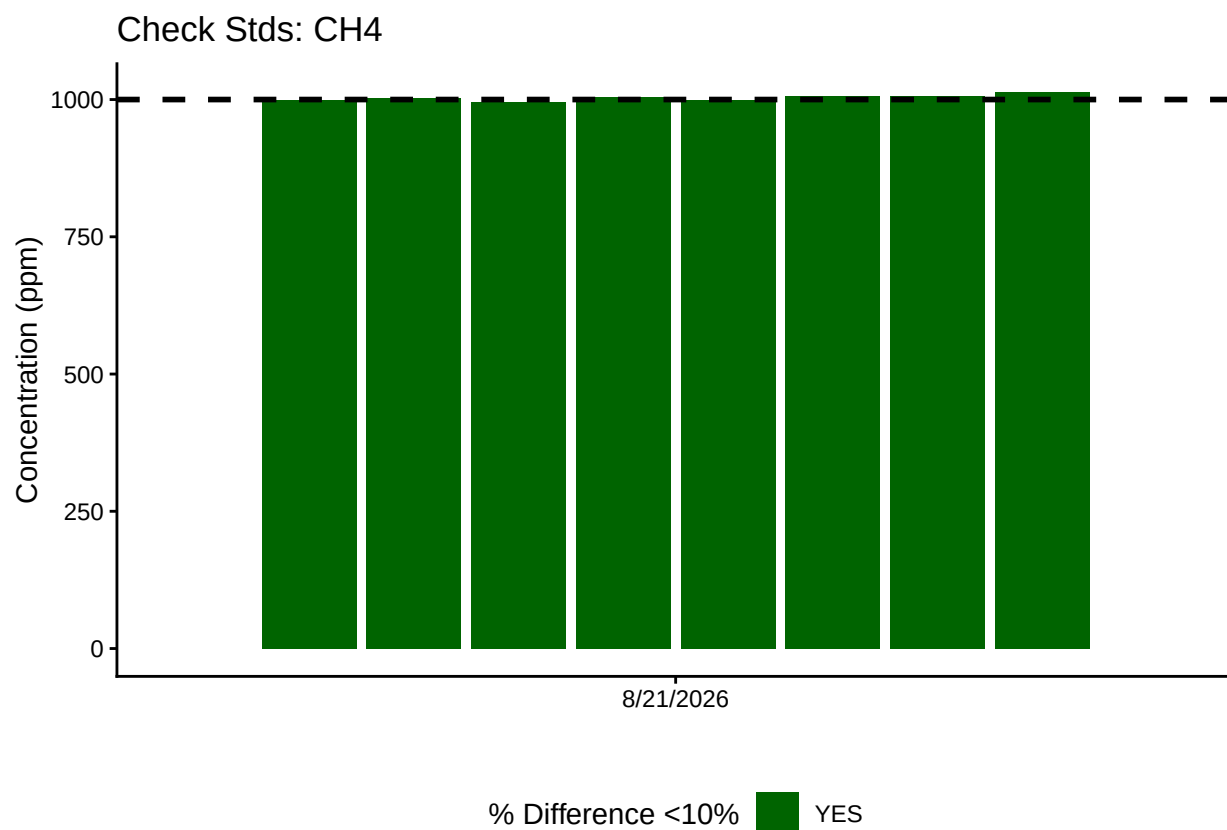
```
## [1] "High_R2 is Good-8/21/2026"
```

```
## [1] "Low_R2 is Good-8/21/2026"
```

2.1 Now calculate the CH₄ in ppm

3 Aanlayze the Check Standards

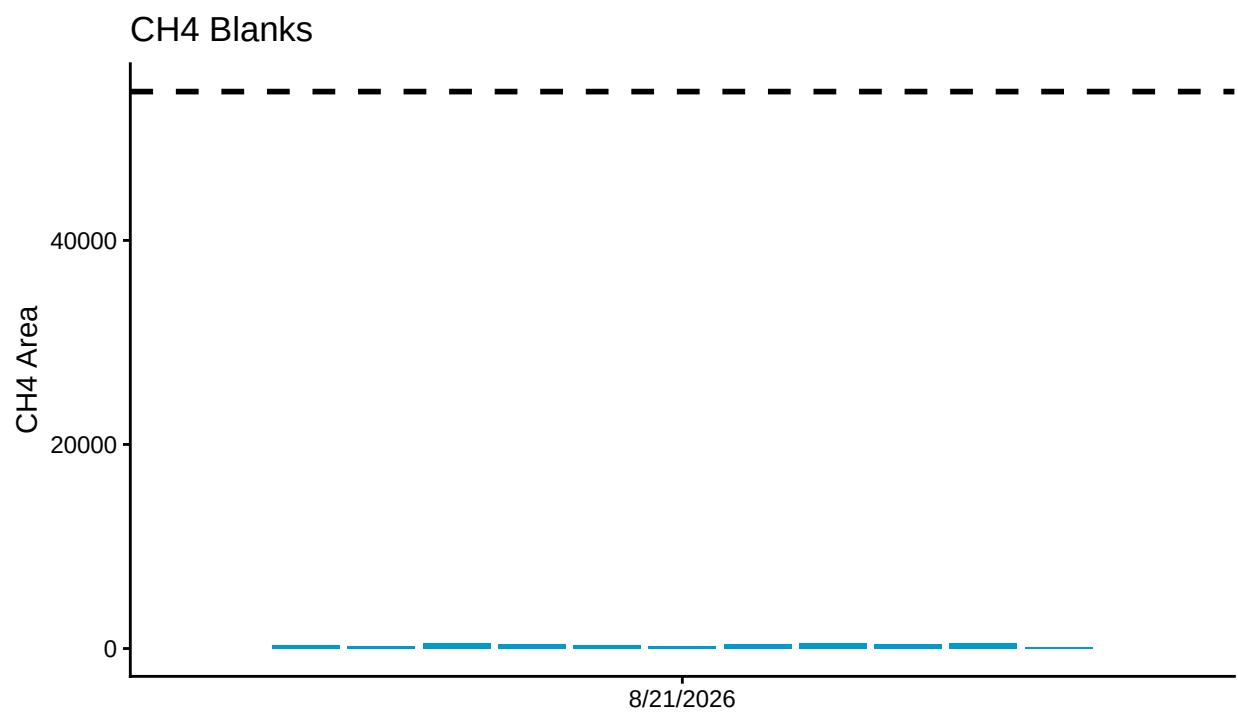
```
## >60% of CH4 Check Standards are within range of expected concentration - PROCEED
```



4 Analyze the Blanks

Assess Blanks

>60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED

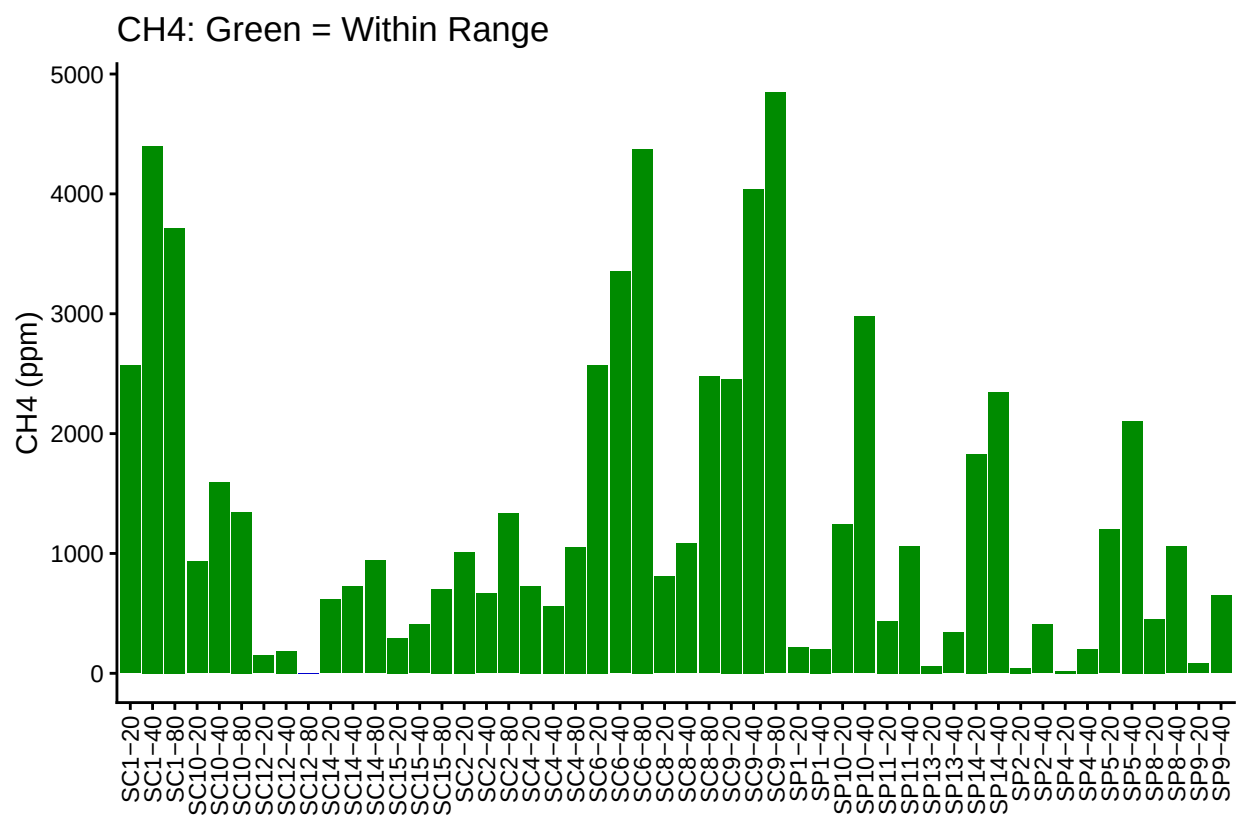
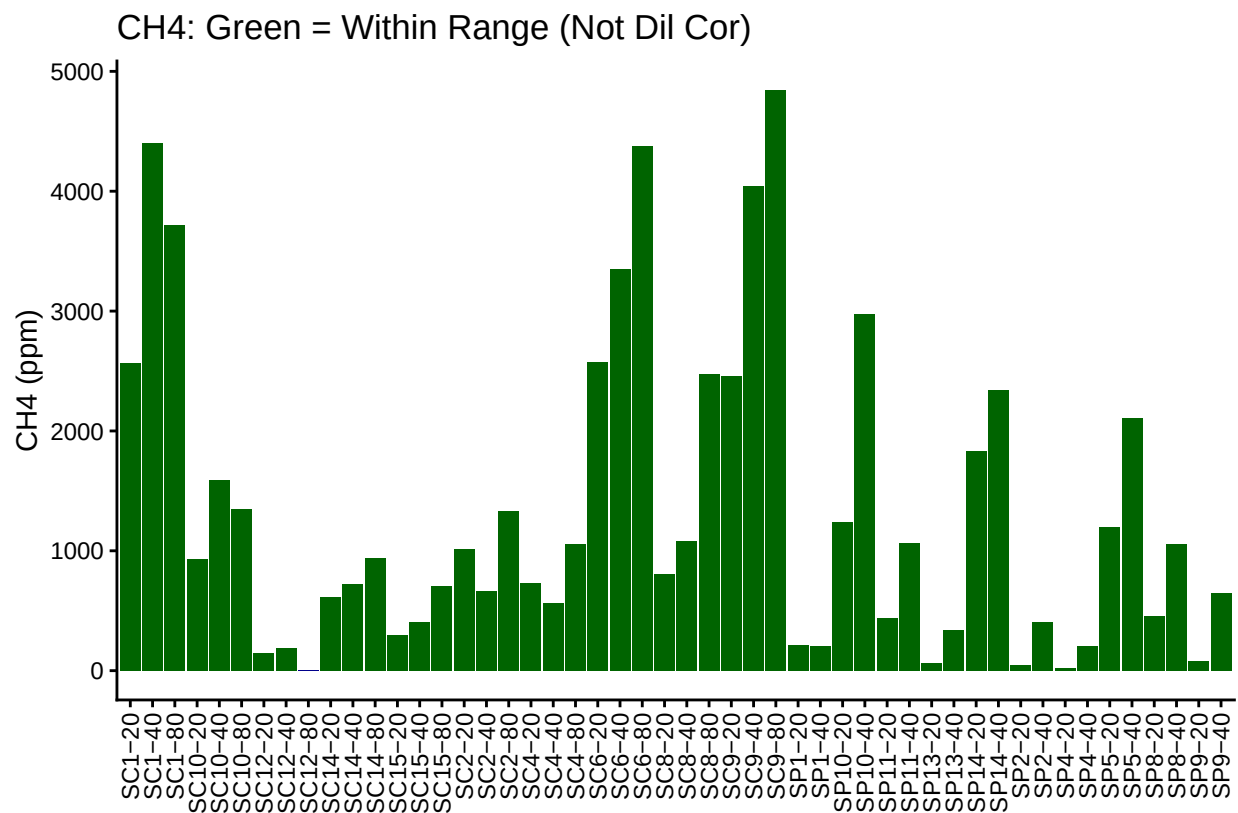


Blank Conc <25% Quartile Samples ■ YES

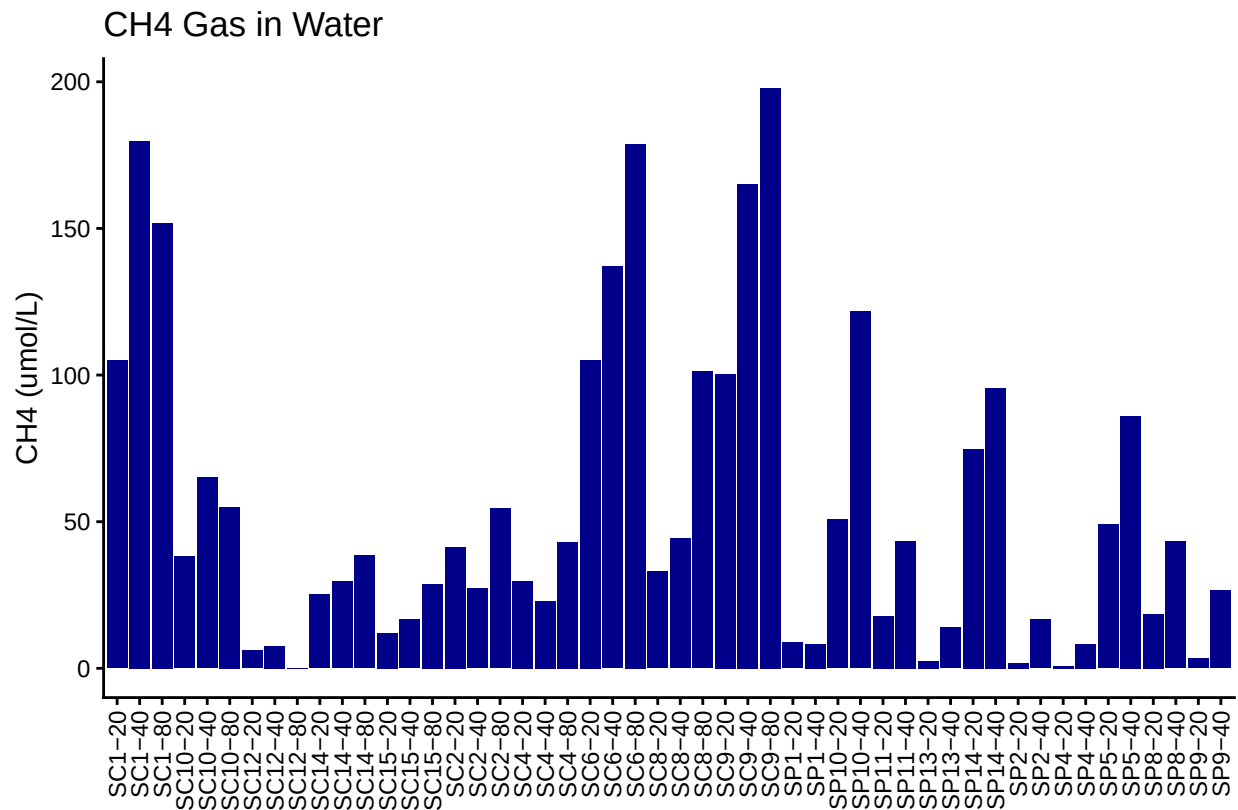
Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	386.2727

5 Dilution correct samples and Check Range



6 Calculate gas in water



7 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

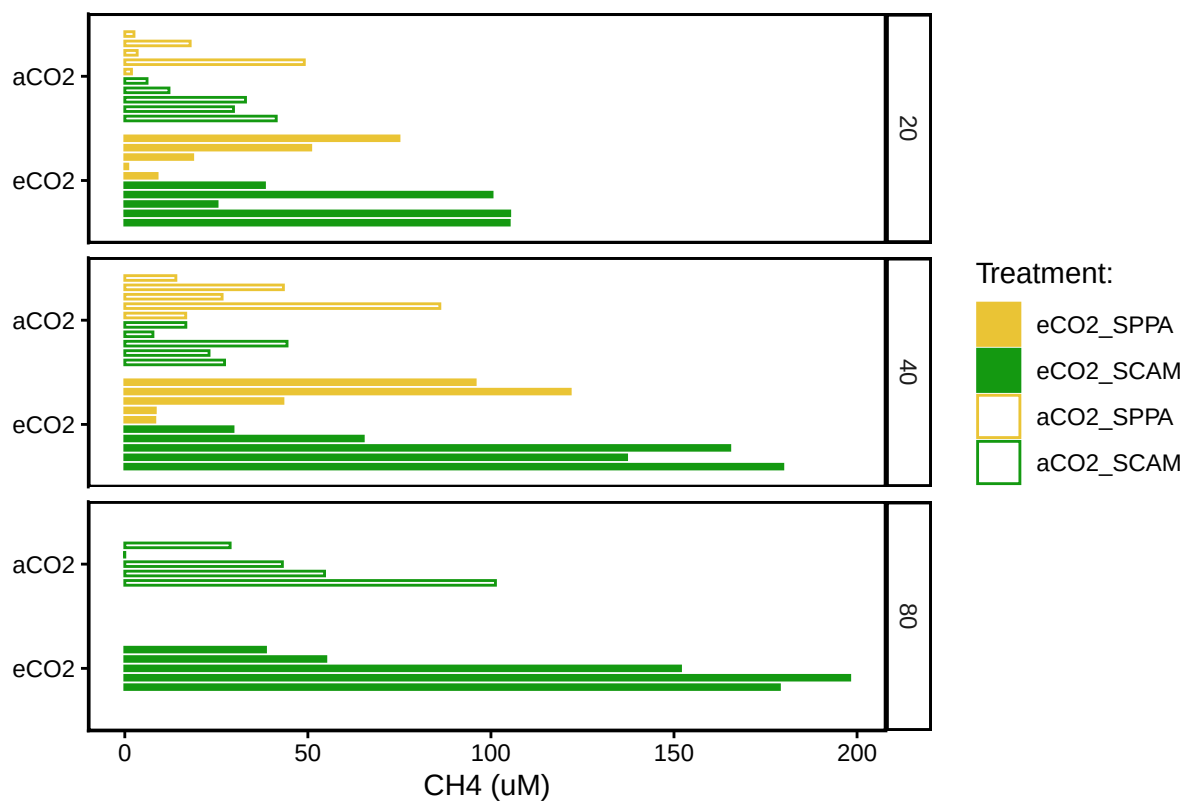
```
## All sample IDs are present in data
```

7.1 Organize Data

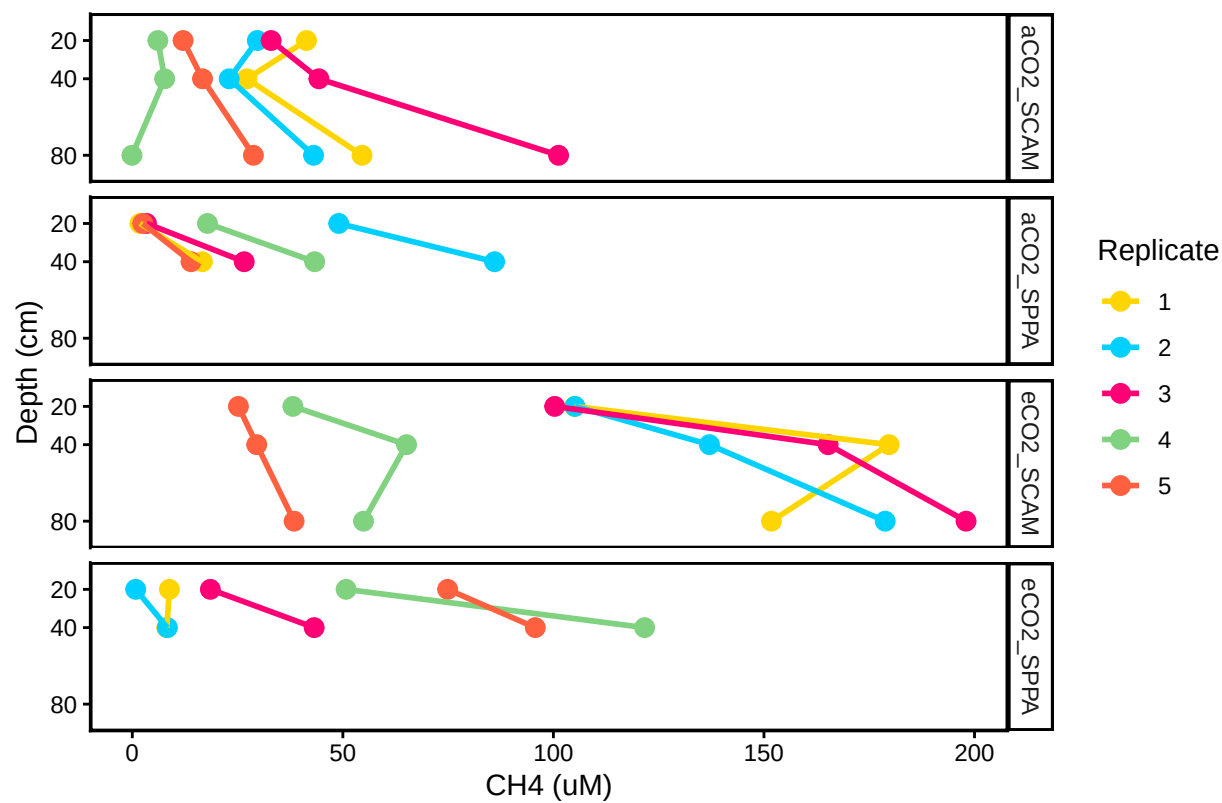
7.2 Visualize Data

```
## Visualize Data
```

CO2xCOMM: Porewater Methane



CO2xCOMM: Porewater CH4 by Replicate



#end